Multi-Line Insurance Policy and Claims Management System

Salesforce-based solution for Vehicle, Property and Life insurance • Condensed Project Report

User Story

A centralized Salesforce solution for managing **Vehicle, Property and Life** insurance policies and claims. Agents create and manage policies through a flexible data model with policy-specific Record Types and Field Sets. Guided Screen Flows simplify quoting, and Apex automation calculates premiums from policy and vehicle details. Claims are routed automatically to the right queue by policy type, and high-value claims go through an approval workflow involving Senior Adjusters and Claims Managers. An LWC Claims Dashboard gives adjusters a focused view of their assigned claims: claim amount, policy type, policy holder, status and days open.

Project at a Glance

Area	What was built
Objects	Policy__c, Claim__c, Contact, User
Record Types	Policy: Auto, Property, Life • Claim: Accident, Property, Life
Automation	AutoQuoting Screen Flow, Claim Routing Flow, Submission Flow, Approver Screen Flow, High Value Claim Approval Process, 17-char VIN Validation Rule
Apex	PremiumCalculator, ClaimsAdjusterController, ClaimsAdjusterControllerTest
LWC	claimsDashboardLwc (Claims Dashboard) with reusable Claim Tile
Security	Permission Sets (Agent, Adjuster, Manager) and state-based Sharing Rules (CA, TX)
Methodology	Agile • 4 sprints • 12 user stories

1. Introduction

1.1 Project Overview

The system centralizes and streamlines management of Vehicle, Property and Life insurance policies and claims, replacing fragmented, manual processes with structured Salesforce objects, Record Types, Field Sets, Flows, Apex automation, Approval Processes and Lightning Web Components.

Agents capture policy data through guided Screen Flows; the system validates it and Apex calculates the premium from factors such as state and model year. When a claim is created, automation identifies the related policy type and routes the claim to the correct queue. High-value claims above the threshold go through a structured approval, and access is controlled with sharing rules and role-specific Permission Sets.

1.2 Purpose

To overcome legacy processes that make quoting, policy issuance and claims handling manual, slow and inconsistent. The solution aims to:

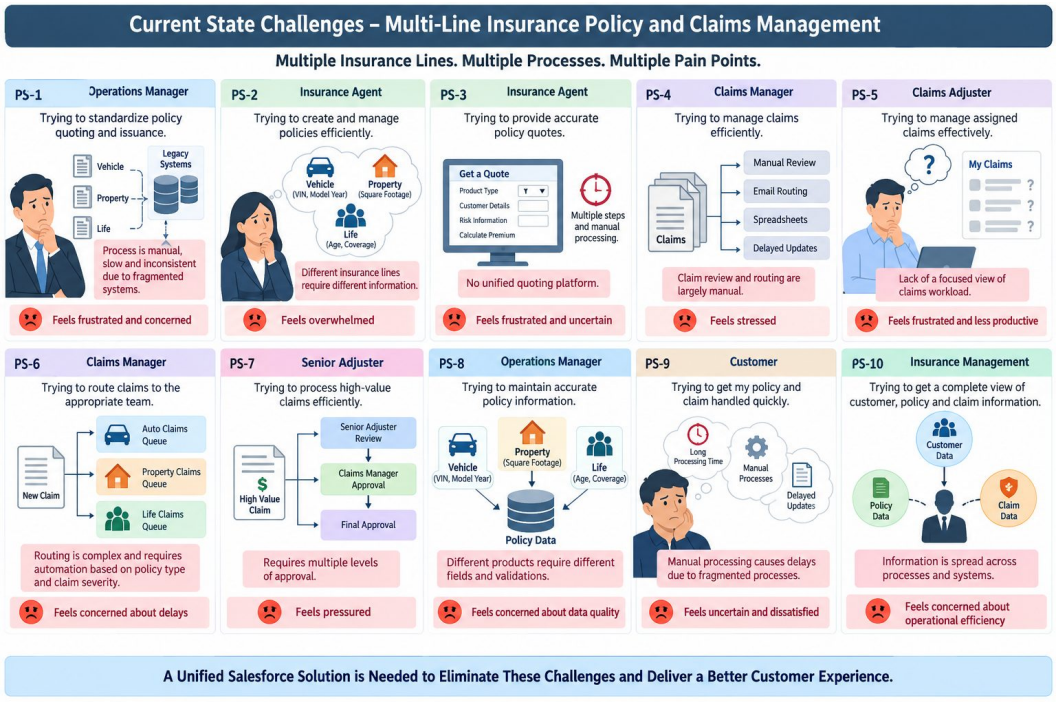
- Standardize and accelerate policy quoting and issuance
- Automate claim initiation and routing by policy type and claim severity
- Provide a 360-degree view of Customer, Policy and Claim information
- Support guided quoting, automated premium and claim processing, and data validation
- Improve claims-adjuster efficiency through an LWC dashboard

2. Ideation Phase

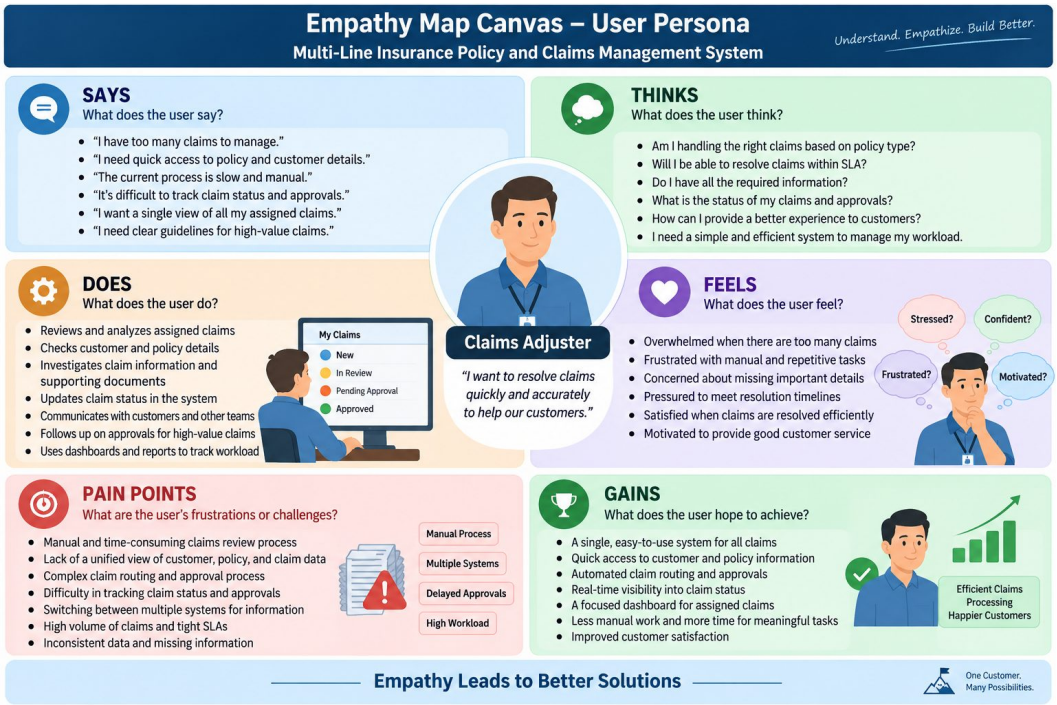
2.1 Problem Statements

#	I am	Trying to	But / Because	Feel
PS-1	Operations Manager	Standardize quoting & issuance	Process is manual and slow; legacy systems are fragmented	Frustrated
PS-2	Insurance Agent	Create & manage policies	Vehicle, Property, Life need different data	Overwhelmed
PS-3	Insurance Agent	Give accurate quotes	Multi-step manual quoting; no unified platform	Uncertain
PS-4	Claims Manager	Manage claims efficiently	Review and routing are manual, not centralized	Stressed
PS-5	Claims Adjuster	Manage assigned claims	No focused view; info not in one place	Less productive
PS-6	Claims Manager	Route claims to right team	Routing by type and severity is complex	Worried about delays
PS-7	Senior Adjuster	Process high-value claims	Needs multi-level structured approval	Pressured
PS-8	Operations Manager	Keep policy data accurate	Each product needs different fields/validations	Concerned
PS-9	Customer	Get policy/claim handled fast	Fragmented manual processing causes delays	Dissatisfied
PS-10	Management	Get complete customer-policy-claim view	Info spread across systems; no unified platform	Concerned

2.2 Current-State Challenges & Empathy Map (Claims Adjuster)



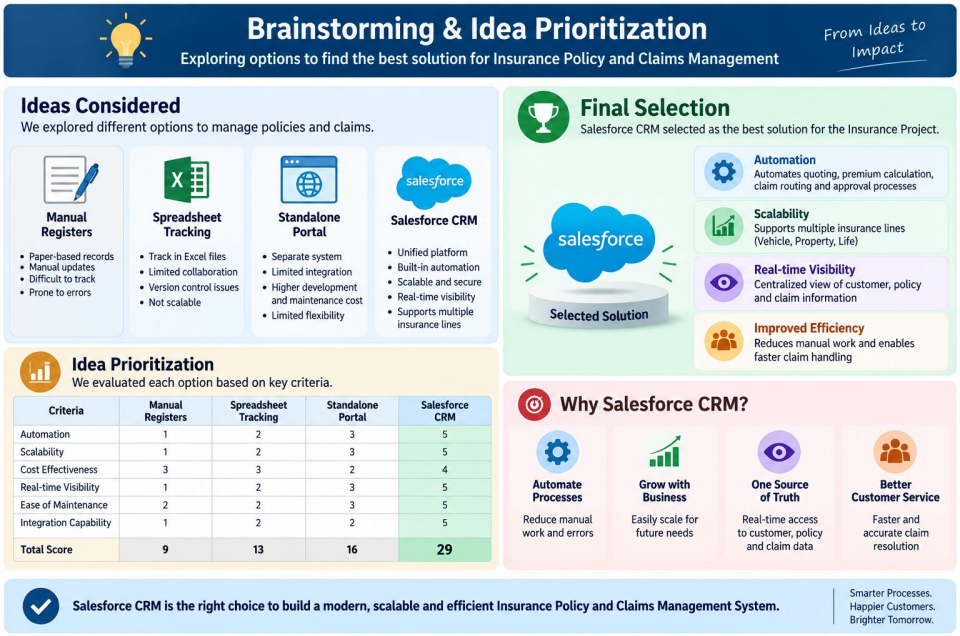
Current-state challenges across personas



Empathy map: Insurance Claims Adjuster

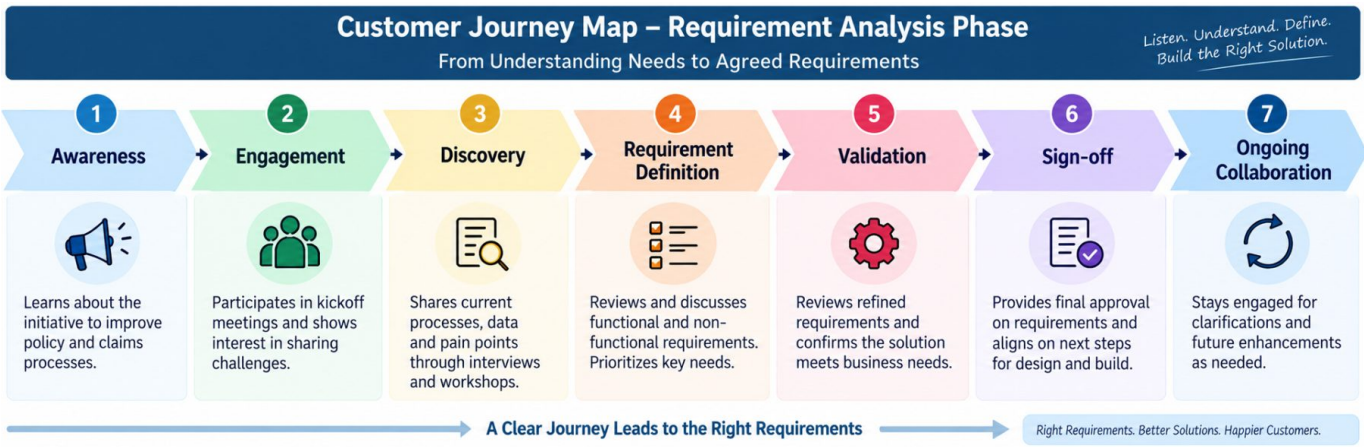
- **Pain:** manual processing, complex routing, limited visibility • **Gain:** claims dashboard, automation, faster approvals

2.3 Brainstorming & Idea Selection



Options considered: Manual Registers, Spreadsheet Tracking, Standalone Portal, **Salesforce CRM (selected)** for automation of quoting/premium/routing/approvals, scalability across insurance lines, real-time visibility, a unified platform and reduced manual effort.

3. Requirement Analysis – Customer Journey



Stages: Inquiry → Policy Selection → Quote Request → Information Capture → Premium Calculation → Policy Issuance → Claim Initiation → Claim Routing → Review & Approval → Resolution → Customer Confirmation → Reporting & Visibility.

3.2 Solution Requirements

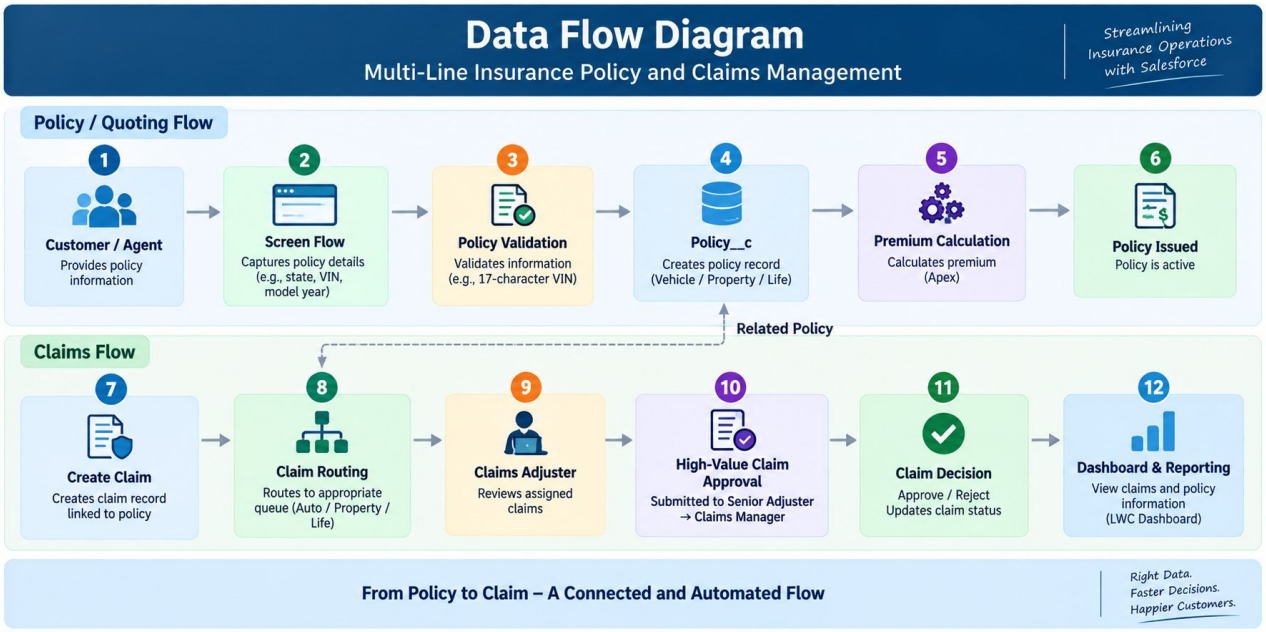
Functional Requirements

ID	Requirement	ID	Requirement
FR-01	Create/manage Policy records for Vehicle, Property, Life	FR-11	Identify high-value claims by amount threshold
FR-02	Separate Record Types for Vehicle, Property, Life	FR-12	Multi-level approval: Senior Adjuster → Claims Manager
FR-03	Product-specific data (VIN; Sq. Ft./Year Built)	FR-13	Approve/Reject with approver comments
FR-04	Guided Auto Quoting Screen Flow (holder, start date, state, VIN, model year)	FR-14	LWC Claims Dashboard of assigned claims
FR-05	Create a draft policy during quoting	FR-15	Show Claim No., Policy Type, Holder, Amount, Status, Days Open
FR-06	Apex-based premium calculation	FR-16	Filter claims: All, Auto, Property, Life
FR-07	VIN must be exactly 17 characters	FR-17	Record-Triggered Flows update related Policy/Claim
FR-08	Create/manage Claims linked to policies	FR-18	State-based access / sharing
FR-09	Claim Record Types: Accident, Property, Life	FR-19	Permission Sets for Agents, Adjusters, Managers
FR-10	Auto-route claims to queue by policy type	FR-20	Reporting on Customer, Policy, Claim data

Non-Functional Requirements

ID	Quality	Expectation
NFR-01	Usability	Simple guided Screen Flows, minimal clicks for agents
NFR-02	Performance	Quoting, premium, routing and dashboard run efficiently
NFR-03	Scalability	Supports 3 insurance lines; room for more products
NFR-04	Data Integrity	Validated, consistent policy and claim data
NFR-05	Security	Access by responsibility: state-based sharing + Permission Sets
NFR-06	Maintainability	Modular Flows, Apex, LWC, Record Types, Field Sets
NFR-07	Reliability	Routing, premium and approvals execute consistently
NFR-08	Extensibility	One data model accommodates product-specific fields
NFR-09	Visibility	Timely view of Customer, Policy, Claim data
NFR-10	User Adoption	Intuitive experience that reduces manual effort

3.3 Data Flow



Policy / quoting flow and claims flow

Flow: Customer/Agent → Screen Flow → Policy Validation → Policy__c → PremiumCalculator → Claim__c → Claim Routing Flow → Claims Queue → Adjuster → Approval Process → Claim Update → LWC Dashboard / Reports

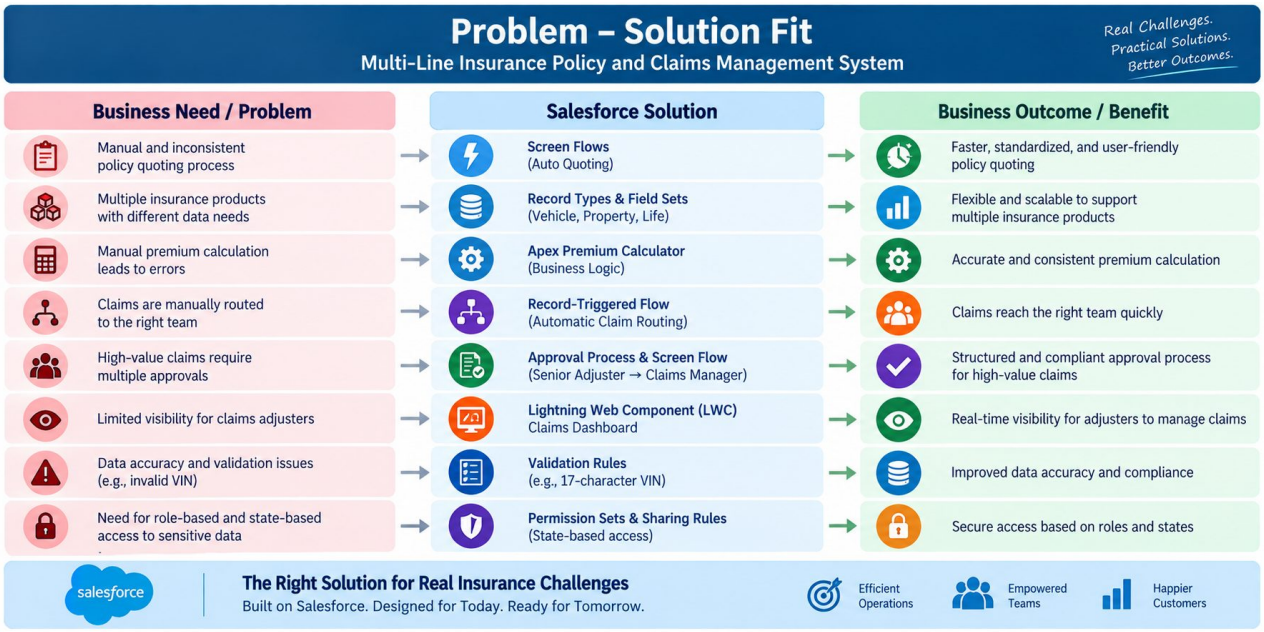
Step	What happens
1–2	Agent provides data; AutoQuoting Screen Flow captures holder, start date, state, VIN, model year
3–4	Validation (17-char VIN) → valid data stored in Policy__c with Vehicle/Property/Life Record Type
5	PremiumCalculator Apex calculates the premium
6–7	Claim__c created and linked to Policy; Record-Triggered Flow routes it to Auto/Property/Life queue
8–10	Adjuster reviews on dashboard; claims above threshold go Senior Adjuster → Manager; Approve/Reject + comments update status
11	Dashboard shows amount, policy type, holder, status, days open

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning, Lightning Web Components
Logic / Processing	Flows, Apex (Premium Calculator, Controllers)
Database	Custom Objects Policy__c, Claim__c; Contact, User
Automation	Record-Triggered Flows, Validation Rules, Approval Processes
Configuration	Record Types, Field Sets, Custom Fields
Security / Reporting / Testing	Permission Sets, Sharing Rules • Reports & Dashboards • Apex Test Classes

4. Project Design Phase

4.1 Problem–Solution Fit

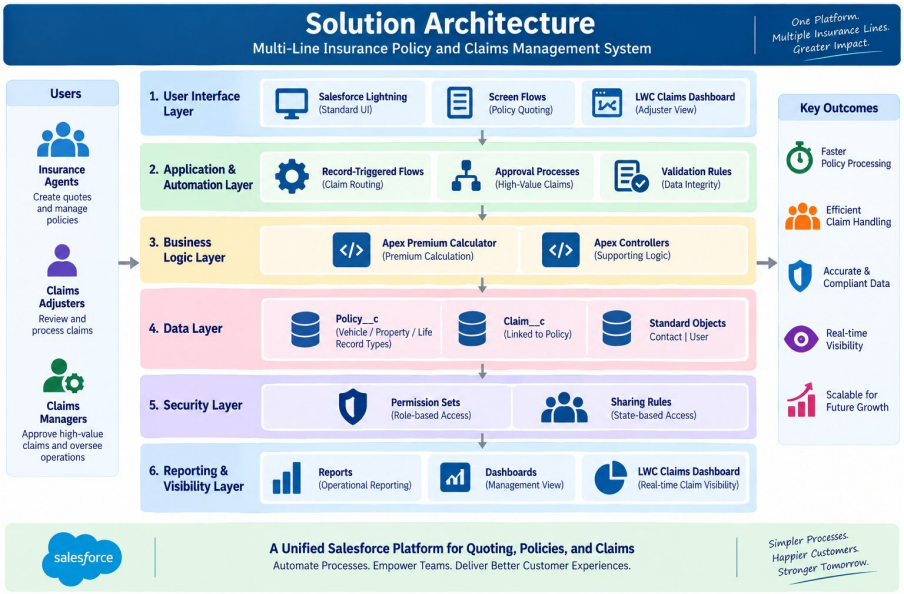


Salesforce directly addresses the key challenges through configurable automation, flexible data management, secure access and real-time visibility.

4.2 Proposed Solution

Parameter	Description
Problem	Manual, inconsistent policy quoting, issuance and claims handling
Solution	Salesforce CRM-based Insurance Policy & Claims Management System
Innovation	Automated quoting, premium calculation, claim routing, high-value claim approval
Impact	Faster policy processing, efficient claim handling, better customer service
Scalability	Flexible data model for multiple products and future expansion
Visibility / Security	Claims Dashboard • role-based and state-based access (Permission Sets, Sharing Rules)

4.3 Solution Architecture



Layer	Implementation
User	Insurance Agents, Claims Adjusters, Claims Managers
UI / Automation	Lightning, Screen Flows, LWC • Record-Triggered Flows, Approval Processes, Validation Rules
Business Logic / Data	Apex Premium Calculator & Claims Controller • Policy & Claim objects with Record Types
Security / Reporting	Permission Sets, state-based Sharing Rules • Claims Dashboard, Reports

5. Project Planning & Scheduling

Agile methodology: Epics → Stories → Story Points, sprint-based execution, velocity used for estimation. Four sprints of 6 days each, planned at 20 story points per sprint.

Sprint	Duration	Start date	Planned end date	Story points
Sprint-1	6 days	24 Oct 2022	29 Oct 2022	20
Sprint-2	6 days	31 Oct 2022	05 Nov 2022	20
Sprint-3	6 days	07 Nov 2022	12 Nov 2022	20
Sprint-4	6 days	14 Nov 2022	19 Nov 2022	20

5.1 Product Backlog

Sprint	Epic	US	User story	Pts
1	Data Modeling	US-01	Admin creates Policy and Claim objects to store policy/claim data	5
1	Data Modeling	US-02	Admin configures Vehicle, Property, Life Record Types on one Policy structure	3
1	Data Modeling	US-03	Admin configures Field Sets and product-specific fields	5
1	Quoting & Automation	US-04	Agent uses guided Auto Quoting Screen Flow	8
2	Validation	US-05	VIN validation during vehicle quoting	3
2	Apex	US-06	PremiumCalculator Apex calculates premiums	8
2	Automation	US-07	Premium calculation integrated with the quoting Flow	5
2	Claims Automation	US-08	Claims routed to support queue by policy type	8
3	Apex	US-09	ClaimsAdjusterController retrieves adjuster's assigned claims	8
3	LWC Dashboard	US-10	Claims dashboard and reusable claim tile	8
4	Approval	US-11	High-value claims through approval process + approver Flow	8
4	Security & Testing	US-12	Tests, sharing rules and Permission Sets	8

Milestones in development: (1) Core Data Model & Policy Configuration • (2) Policy Issuance & Claim Routing Automation • (3) Claims Adjuster LWC Dashboard • (4) Advanced Claim Processing, Security & Code Setup. Detailed implementation follows.

6. Development Phase – Milestone 1: Core Data Model & Policy Configuration

Policy object (Policy__c)

- Record name **Policy Number**, Auto Number format **P-{0000}** (start 1); reports, activities and field history enabled; deployed and searchable.
- Record Types: **Auto, Property, Life**. Field Sets: Vehicle (VIN, Model Year), Property (Square Footage, Year Built), Life (Beneficiary Name, Policy Term Months).

Policy and Claim fields

Policy field	Type	Claim field	Type
Beneficiary Name	Text(50)	Adjuster	Lookup(User)
Customer	Lookup(Contact)	Approval Status	Picklist: New, Submitted for Approval, Approved, Rejected
Model Year / Year Built	Text(5)	Claim Amount	Currency(16,2)
Policy Start Date	Date	Date of Loss (required)	Date/Time
Policy Term Months / Square Footage	Number(18,0)	Description	Long Text Area
Premium	Currency(16,2)	Policy	Lookup(Policy)
VIN	Text(20)		

Claim object (Claim__c)

- Record name **Claim Number**, Auto Number **C-{0000}**; Record Types: **Accident, Property, Life**; reports and field history enabled.

AutoQuotingFlow (Screen Flow)

- **Get Records** – Policy Record Type = Auto (Get Vehicle RT ID).
- **Screen 1** – Policy Holder (Contact lookup), Policy Start Date, Policy State (picklist).
- **Screen 2** – Vehicle details: VIN and Model Year.
- **Create Records – Create Draft Policy** maps holder, start date, state, VIN, model year and Record Type; the new Id is stored in `varPolicyId`. Flow saved and activated as **AutoQuotingFlow**.

Validation Rule: VIN_Must_Be_17_Characters

```
AND( RecordType.DeveloperName = "Auto", LEN( VIN__c ) <> 17 )
Error message: The Vehicle Identification Number (VIN) must be exactly 17 characters long for Auto Policies.
Error location: VIN field
```

6. Milestone 2: Complex Policy Issuance & Claim Routing Automation

PremiumCalculator (Invocable Apex)

Accepts policy Ids, queries VIN, Model Year and State, and returns the premium for each policy:

```
@InvocableMethod(label='Calculate Policy Premium')
public static List<Decimal> calculatePremium(List<PremiumInputWrapper> inputs) {
    // collect policyIds, then query the policies:
    // SELECT Id, VIN__c, Model_Year__c, Policy_State__c FROM Policy__c WHERE Id IN :policyIds
    for (Policy__c policy : policiesToRate) {
        Decimal basePremium = 1000.00;
        if (policy.Policy_State__c == 'CA')      basePremium *= 1.15;
        else if (policy.Policy_State__c == 'TX') basePremium *= 1.05;
        if (policy.Model_Year__c != null && Integer.valueOf(policy.Model_Year__c) < 2018)
            basePremium += 100;
        premiumResults.add(basePremium);
    }
    return premiumResults;
}
```

Integrating premium into the quoting Flow

- Inside **AutoQuotingFlow**, after Create Draft Policy, an **Action** element **Calculate Premium Action** calls PremiumCalculator with `varPolicyId`; the output is stored in `varCalculatedPremium` (Currency, 2 decimals).
- An **Update Records** element (*Update Policy With Premium*) finds the policy where Record Id = varPolicyId and sets **Premium** = varCalculatedPremium. Saved as a new version and activated.

Claim Routing – Record-Triggered Flow (After Save on Claim)

Element	Configuration
Trigger	Claim object, when a record is **created**; optimized for Actions and Related Records
Get_Policy_RT	Get the related Policy (Record Id = triggering Claim → Policy); store Record Type in `varPolicyRTName`
Decision: Route by Policy Type	Outcome 1 **Auto Claim Route** (varPolicyRTName = Auto) • Outcome 2 **Property Claim Route** (= Property) • Default **Life Claim Route**
Update Records	**Assign to Auto Queue**, **Assign to Property Queue**, **Assign to Life Queue**: set the Claim **Owner Id** to the matching queue Id

6. Milestone 3: Claims Adjuster LWC Dashboard

Apex: ClaimsAdjusterController

A `ClaimWrapper` class exposes *claimId, claimNumber, claimAmount, status, policyType, policyHolderName, daysOpen*. The cacheable `@AuraEnabled getAssignedClaims()` method returns all claims owned by the logged-in user, joined with Policy and Customer data:

```
List<Claim__c> claims = [
    SELECT Id, Name, Claim_Amount__c, Approval_Status__c, Date_of_Loss__c,
           Policy__r.Name, Policy__r.RecordType.DeveloperName,
           Policy__r.Customer__r.FirstName, Policy__r.Customer__r.LastName
    FROM Claim__c
    WHERE OwnerId = :UserInfo.getUserId()
    ORDER BY Date_of_Loss__c DESC ];
for (Claim__c claim : claims) { results.add(new ClaimWrapper(claim, claim.Policy__r, customer)); }
return results;
```

LWC: claimsDashboardLwc

- Created as a Lightning Web Component, exposed (*isExposed*) for **Lightning App page, Home page and Record page**.
- HTML: a **lightning-card** titled “My Assigned Claims Dashboard” with a **lightning-combobox** filter (**All, Auto, Property, Life**) and a responsive grid that loops over visible claims.
- JS: the Apex method is called with **@wire**; *allClaims* holds the result and *visibleClaims* is filtered client-side when the picklist changes; errors are captured in *error*.
- **Claim Tile** child component (reusable): shows claim number, amount, status, policy holder and days open, with an icon by policy type (e.g. truck for Auto, home for Property).

6. Milestone 4: Advanced Claim Processing, Security & Code Setup

Approval Process: High Value Claim Approval (on Claim)

Setting	Value
Entry criteria	**Claim Amount > 50,000**
Record editability	Administrators only while in approval
Approval page layout	Claim Number, Owner, Claim Amount, Adjuster, Policy
Initial submitters	Claim owner
Step 1 – Senior Adjuster	Auto-assigned to the Senior Adjuster user; approve/reject by first response
Step 2 – Manager Review	Auto-assigned to Department Manager; on rejection, perform all rejection actions for the step and final rejection actions
Field updates	Initial submission → Approval Status = Submitted for Approval • Final approval → Approved • Final rejection → Rejected

Submission Automation Flow (Record-Triggered, After Save)

- Runs when a Claim is created/updated with **Claim Amount > 50000** and **Approval Status = New**.
- Action **Submit Claim to Approval**: Record Id = triggering claim, Approval Process = *High Value Claim Approval*.

Claim Approver Screen Flow + Quick Action

- Input variable `recordId` → **Get Claim Details** → **Claim Review Screen** showing Claim ID, Amount and Policy Type, with **Approve / Reject** radio buttons and a long-text **Approver Comments** field.
- **Check Decision**: Approved path runs **Approve Claim** action; default path runs **Reject Claim** action, both passing the comments.
- Saved and activated as **Claim Approver Screen Flow**, then exposed as the Claim quick action **Approve/Reject Claim** and added to the Lightning Experience actions on the Claim page layout.

Claim Policy Holder State Update Flow

- Record-Triggered Flow on Claim: gets Policy records, loops through them and copies the policy state into the claim's **Policy Account Holder State** field, which the sharing rules use.

Apex Test Class: ClaimsAdjusterControllerTest (target 95%+ coverage)

- `@testSetup` creates a Standard User, a Contact (John Doe), an Auto Policy with a VIN and an Accident Claim (amount 5,000, loss date 10 days ago), all owned by the test user.
- `testGetAssignedClaims` runs as that user and asserts claimId/claimNumber are not null, status is **New**, amount is 5000, holder name contains "John" and policy type is not "Unknown".

Sharing Rules (Claim object)

Rule	Criteria	Access
Share Records based on Policy State CA	Policy Account Holder State equals CA	Read/Write
Share Records based on Policy State (TX)	Policy Account Holder State equals TX	Read/Write

Permission Sets

Permission Set	Object access	System / Apex
Insurance Agent Access (quoting only)	Policy: Create, Read • Contact: Read, Edit • Claim: Read	Flow User enabled; PremiumCalculator Apex enabled
Claims Adjuster Access (claim handling)	Policy: Read • Contact: Read, Edit • Claim: Read, Edit	ClaimsAdjusterController Apex enabled
Claims Manager Access (reporting & approvals)	Policy: Read • Contact: Read, Edit • Claim: Read, Edit, Delete	Run Reports; Manage Approvals

7. Functional & Performance Testing

- **Data model:** Policy and Claim records created with Auto / Property / Life Record Types and Field Sets.
- **Quoting:** AutoQuotingFlow creates a draft policy; a VIN that is not 17 characters is rejected by the validation rule; premium is calculated and written to the Premium field.
- **Routing:** a new Claim is assigned to the Auto, Property or Life queue according to its policy's Record Type.
- **Approval:** a claim above 50,000 is submitted automatically and moves Senior Adjuster → Manager; Approval Status updates to Submitted for Approval / Approved / Rejected.
- **Dashboard & Apex:** LWC shows the adjuster's claims with working All/Auto/Property/Life filter; ClaimsAdjusterControllerTest passes with high coverage.
- **Security:** state-based sharing (CA, TX) and role-specific Permission Sets verified for Agents, Adjusters and Managers.

8. Advantages & Disadvantages

Advantages	Disadvantages
Automated claim routing by policy type	**Complex data management** – each policy line needs different fields
Faster processing – approvals and escalation reduce manual effort	**Approval complexity** – amount, policy type and adjuster level
Better visibility – LWC dashboard for adjusters	**User adoption** – agents need intuitive flows with few clicks
Data accuracy – validation rules	**Integration complexity** – external callouts need extra Apex and mock tests
Flexible model – Record Types and Field Sets	**Salesforce dependency** on platform configuration
Secure access – Permission Sets and Sharing Rules	**Implementation effort** – many Flows, Apex, LWC and rules to maintain
Scalable to more lines and enhancements	

9. Conclusion

The Multi-Line Insurance Policy and Claims Management System provides a centralized Salesforce-based solution for managing Vehicle, Property and Life insurance policies and claims. It replaces manual, fragmented processes with guided quoting, automated premium calculation, claim routing, a structured approval process and an LWC-based Claims Dashboard.

By combining Flows, Apex, Record Types, Field Sets, Validation Rules, Approval Processes, LWC, Permission Sets and Sharing Rules, the project improves policy processing, claim handling, data accuracy, security and operational visibility, showing how Salesforce can bring multiple insurance processes onto a single scalable platform.

10. Future Scope

Area	Enhancement
External System Integration	Real-time quote verification with external vehicle and property databases
Additional Insurance Products	Extend beyond Vehicle, Property and Life
Advanced Claims Automation	Automated claim assessment and escalation based on claim characteristics
Enhanced Analytics	Advanced dashboards and reports for policy, premium and claims analysis
Customer Experience	Customer-facing policy and claim status tracking
Mobile Accessibility	Key Salesforce processes available on mobile devices
AI-Based Assistance	AI help for agents and adjusters on policy and claims tasks
Integration Architecture	Expanded API-based integrations with external verification systems

11. Appendix

Category	Items
Objects	Policy, Claim, Contact, User
Record Types	Vehicle, Property, Life; Accident, Property, Life
Automation	Screen Flows, Record-Triggered Flows, Approval Processes
Apex	PremiumCalculator, ClaimsAdjusterController
LWC	Claims Dashboard, Claim Tile
Validation / Security	17-character VIN validation • Permission Sets, State-Based Sharing Rules
Testing / Reporting	ClaimsAdjusterControllerTest • Claims Dashboard & Salesforce Reports/Dashboards

Thank You